

ISDN

Module 27

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0:00 [Music]

0:16 hello viewers I welcome you to the lecture Series in cyber security and information security today we will

0:23 discuss ISDN security ISDN that is integrated Services digital Network is a method to transfer voice data with some

0:32 particular Services accessible for data ISDN is a set of CCI /it standards for circuit switch

0:41 transmission of data over various media using ordinary telephone grade copper wire ISDN provide worldwide digital

0:49 communication service in the transition to electronic documents and business transactions ISDN digital nature

0:57 facilitates adding security but it has been deployed with a little thought to security ISDN provides digital circuit

1:05 switched voice and data services as well as packet switch data services to users

1:12 ISDN is a method to transfer voice data with some particular Services accessible for data history of

1:21 ISDN before the coming of age of ISDN original analogy telecommunication Services also known as plain old

1:30 telephone system PS were the only source of telecommunication worldwide PS was the originally acronymic for post office

1:39 telephone service system but changed when the post offices stopped offering telephone services PS comprised mainly

1:47 of copper wires which link the subscribers home to the central switch office this form of telecommunication

1:54 had its limitations longdistance calls had to be routed through operators and switchboards which made them highly

2:02 unreliable and timeconsuming there was also the issue of static interference known as the line noise which Disturbed

2:10 communication in the 1960s the telecommunication industry began working on the converting its analog systems to

2:18 transmit packets of digitized data by digital switching the United Nations International Telephone and Telegraph

2:25 consultative committee now known as International telecommunication Union telecommunication iut actively pushed

2:34 and encouraged the research for integrated Services digital Network by starting a movement to recommend and provide Sands for international

2:43 digitization of the telecommunication services this was initiated in 1984 two major US networks Northern

2:52 telecommunication and atnt took the first steps towards implementing ISDN but did not interoperate with existing

2:59 equipments and software available to telecommunication networks this led to a setback for ISDN worldwide by 1990 due

3:09 to the worldwide effort National ISDN 1 n11 was made compatible with existing

3:16 proprietary equipments for telecommunication services this way people did not need to switch brand or buy software for the network they set

3:26 the standards and procedures for future digital telecommunication technology to be used by everyone ISDN has resulted in better

3:35 voice quality and internet access due to its packet switch connection with ISDN voice and data are carried by a bearer

3:44 channel that is B Channel with a bandwidth of 64 kbps sometimes 56 kbps

3:52 as opposed to telephone line which carry 52 kbps a data channel that is D channel

3:59 is used for controlling Network Services and signaling to construct bearer connections and carry data over the bearer

4:07 channels this carries a bandwidth of 16 kbps or 64 kbps for now the future of

4:15 ISDN lies with Broadband ISDN that is B ISDN this entails transferring data

4:22 voice and video all at the same time using fiber optic telephone line which can carry data rates from 155 Mbps to

4:32 622 Mbps and Beyond this is the major topic for research and development now

4:39 we come to the types of ISDN there are two types of ISDN first basic rate ISDN

4:47 that is B second primary rate ISDN that is p there are two services offered by ISDN

4:57 this is known as the B and b1 b1rri basic rate interface consists of 2 bearer

5:05 channels and 1 16 kbps data Channel totaling 144 kbps as the name implies B

5:14 usually has enough bandwidth for individual users PRI primary rate interface is used for clients that

5:22 require greater bandwidth it usually comprised of 2 364 kbps B channels and

5:29 one 164 kbps D Channel totaling 1.536

5:36 Mbps sometimes the pri service can have 30 B channels and One D channel for a total of

1.98 4 Mbps brri has two Bearer that is B channels for transfer data which operate at 64 kbps each the other channel is known as the Delta D Channel and operates at 6 16 kbps the T channel is used for link management purposes and signaling this is known as 2B plus 1D PR depends on the country but a typical PR is 23 B channels that is B channels at 64 kbps for data and One D Channel at 64 kbps one format of the pr interface consisted of 24 DS zeros and looked a lot like channelized T1 circuit the PRI is similarly defined for international use where the fundamental historical transport bit rate is 2.48 Mbps sometimes called E1 in this case one of the 32 DS Z was already used for control purpose so there were only 31 available DS zeros for ISDN hence the format was 30b Plus D the major difference between a garden variety T1 and a PR is that the P normally has 23 channels used for transport of digital voice or data conversations called Bearer or B channels while the 24th channel was used for control called the data or D Channel thus a PRI ISDN circuits is often referred to as 23b plus d normally is inserted above because One D channel can control more than 23 B channels so it was also possible to have a 24b ISDN circuit so long as the control came from another PR so now we come to the features of isgn the major feature of isgn is that it put together speech and information on the same line which were not presented in the classic telephone system ISDN services are ability to deliver maximum data within the combination of voice data video facts over a single line while it provide at least two instantaneous connections ISDN make available access to packet switch networks because it is a circuit switch telephone Network system therefore it provides better quality of voice data than a analog phone user can attach several devices to the line according to their need instead of purchasing many analog phone lines the ISDN B channel is responsible to provide greater data rate so now we move on to the ISDN Services ISDN provide three types of services beer Services T services supplementary Services Telly and supplementary Services represent the type of features and functions which are visible to end users while beer Services represent the parts of the network which remain hidden from end users beer Services facilitate the realtime communication of digital information

9:07 between end users these Services main relate to network functions and account for OSI layers 1 to 3 an example of the

9:16 beer Services is that 64 kbps 8 KZ structured speed service this Services

9:24 uses a data rate of 64 kbps together with 8 khz timing information which

9:31 structures the data into octet intervals for transmitting a pulse code modulated that is PCM speech signal the fact that

9:40 the signal represents speech is known to the network allowing it to employ Transformations which may not preserve

9:47 bit Integrity but will result in good quality audio reproduction T services provide a set of higher level functions

9:56 on the top of beerer services these services account for OSI layers 4 to 7

10:03 examples of teleservices are telephony Services which provide speech communication over a b Channel with

10:10 control signaling over the D channel f similar Services which facilitate the communication of bitmap images over a b

10:19 Channel with controlling signals over the D Channel tetch Services which facilitate the interchange and

10:27 communication of textual as well as formatted documents over a b Channel with control signaling over the D

10:34 Channel supplementary Services enhance beerer and T services in an independent fashion examples of supplementary

10:42 services are the Centre service emulates a private Network and provides a specialized feature to a set of

10:51 subscribers The call transfer service allows a user to transfer an active call to a third party the call waiting

10:59 service allows a user already engaged in a call to the informed of another incoming call the calling line ID

11:07 service provides the calling parties address information to the called party although these Services all appear

11:15 geared towards circuit switch telephone calls they are equally applicable to the packet switch data calls working of ISDN

11:24 how does ISDN work one single transmission channel for communication is available in analog Network or

11:32 regular telephone lines that are provided to us by telephone companies therefore only one service can be

11:39 carried at a time that is voice data video at same time whereas in ISDN lines

11:47 there are the same pair telephone lines are logically divided into multiple channels they are normally divided into two

11:55 channels B channel is the first type of Channel 6 4 kbps of data can be transferred using B Channel normally

12:04 ISDN lines possess two B channels one channel is dedicated for voice and the other is dedicated for data

12:13 communication the data transmission takes place on one pair copper wire second type of channel is used for line

12:21 and calling system setup is known as D Channel or Delta Channel also third

12:28 Channel on the other hand have 16 kbps of bandwidth concept of ISDN line

12:35 working with the help of image is as shown ISDN protocols components and

12:42 routers option te1 that is terminal equipment type one this is the ISDN

12:49 telephone computer ISDN fax machine or whatever it is that you have hooked up to the ISDN phone line te2 that is

12:59 terminal equipment type two this is the old analog telephone old style fax machine modem or whatever you use to

13:07 hook up to the analog phone line it can also be other Communications equipment that is handled by a ta ta that is

13:17 terminal adapter this lets old te2 stuff talk to the isdn network it also adapts

13:25 other kinds of equipment like ethernet interfaces to ISDN nt1 that is Network

13:33 termination type one this is the end of the line for the local phone company and the beginning of your house phone

13:42 network nt2 that is Network termination type two

13:49 in most homes this won't exist if you were a big company with your own private telephone system then this would be the

13:56 guts of that telephone system LT that is line termination this is the physical

14:04 connection of the telephone company ET exchange termination this is the local phone

14:12 company's logical connection from your phones to the phone network letters r s

14:19 Tu and V in the diagram are reference point that everyone uses to talk about

14:26 each of these parts of the network the the r reference point is the interface between an old style telephone and

14:34 terminal adapter equipment since most homes won't have any N2 equipment the s

14:41 and t reference points are usually one and the same and are sometimes called St

14:48 bus different things happen in the different parts of the network different wiring requirements different data

14:54 speeds different encoding Etc the reference Point V and LT and ET

15:02 equipment are in the phone company's domain ISDN provides for digital transmission over ordinary telephone

15:10 copper wire as well as over other media uses asdn uses circuit switching to establish a physical permanent Point

15:19 topoint connection from the source to the destination ISDN has standards defined by the itu that Encompass the

15:28 OSI the bottom three layers of which are physical data link and network as shown
15:34 in the diagram difference between ADSL and ISDN there are some obvious differences
15:41 between the two Services ISDN lines provide two voice channels or one 128
15:49 kbps data Channel whereas ADSL is only data line which is used for only data
15:56 the part for ADSL is by a phone company through the medium of copper while which keeps the line stays alive and works
16:05 even when local power fails whereas in ISDN lines power is required locally and does not work if
16:13 the local power fails ISDN advantages the basic advantage of ISDN
16:20 is to facilitate the user with multiple digital channels these channels can operate concurrently through the same one copper
16:29 wire pair the digital signals broadcasting transversely the telephone lines ISDN
16:37 provides High data rate because of digital scheme which is 56 kbps BR ISDN
16:44 using a channel aggregation protocol such as bonding or multi-link triple P supports an uncompressed data transfer
16:53 speed of 128 kbps plus bandwidth for overhead and signaling PR transfers at
17:01 17 minutes, 1 second an even higher speed of up to 1 1920 kbps ISDN allows multiple devices to
17:09 share a single line it is possible to combine many different Digital Data sources and have the information routed
17:17 to the proper destination ISD and network lines are able to switch many fold devices on the
17:24 single line such as faxes computers cash register credit card readers and many
17:30 other devices these all devices can work together and directly be connected to the single
17:37 line ISDN takes only 2 seconds to launch a connection while other modems take 30 to 60 second for
17:46 establishment instead of the phone company sending a ring voltage signal to the ring the bell in your phone inband
17:55 signal it sends a digital packet on a separate Channel that is out of band signal the outof band signal does not
18:03 disturb established connections no bandwidth is taken from the data channels and call setup time is very
18:10 fast as said earlier ISDN disadvantages there are some disadvantages associated with ISDN and
18:18 they are as follows the disadvantage of ISDN lines is that it is very costly
18:25 than the other typical telephone system ISDN line are more expensive when compared to the regular landline telephone

18:33 system ISDN provider companies and ISDN users are required to have special dedicated line and then can Encore at

18:42 extra cost ISDN requires specialized Digital Services just like telephone company so now we come to the security

18:51 issues which is more secure ISDN cable modem DSL T1 Etc on the physic IAL

18:59 medium part EES dropping on a phone land line where dialup modems are used as physically easier than spying on the

19:07 more modern mediums because the lower data rate more easily accommodates homemade Electronics of questionable

19:15 quality also some of the protocols used over various mediums include encryption which may hinder spying attempts on the

19:24 physical line alth the marrow adding in deserted Courtyards during Moon moonless nights crouching under Services boxes to

19:32 our tapping wires is a time honored spying tradition I believe that most attacks nowadays will be on a logical

19:39 layer the attacker will try to hack into the ISP systems or the machines on the customer side be it a cable DSL modem or

19:50 a not totally updated desktop systems and he will do so by sending IP packets from far away these are mostly

19:59 orthogonal to the question of what medium is used logic attacks allow the attacker to try his luck on millions of

20:07 targets from the comfort of his own basement instead of confronting rainy Gloom and ill-tempered cats on the

20:15 random chance that the dozen or so homes that he will approach de night may contain something valuable

20:22 electronically speaking and that no board Insomniac will spot him and call the police physical spying still exist

20:30 but is mostly employed on the specific targets the attacker does not want to spy on anybody he is after you

20:38 personally at which point you should have more pressing needs proofing a some home network against a dedicated

20:45 attacker is hard think about Tempest or even about looking at your display and or keyboard over your shoulder from the

20:53 outside through a window a good telescope a video camera with High FPS and you can record a typed password from

21:02 about 200 M away so for the specifics of DSL T1 cable ISDN I would advise

21:11 selecting the provider choose an ISP with a good reputre about security issues in particular with regards to whatever

21:19 modem he provides that's more important security wise than the physical medium

21:26 and to abstract the whole question away never do sensitive stuff over the Internet unless you employ appropriate

21:33 logical protection such as https so this was all about ISDN security thanks for watching this

21:41 program thank you

21:44 [Music]